MILITARY STANDARD

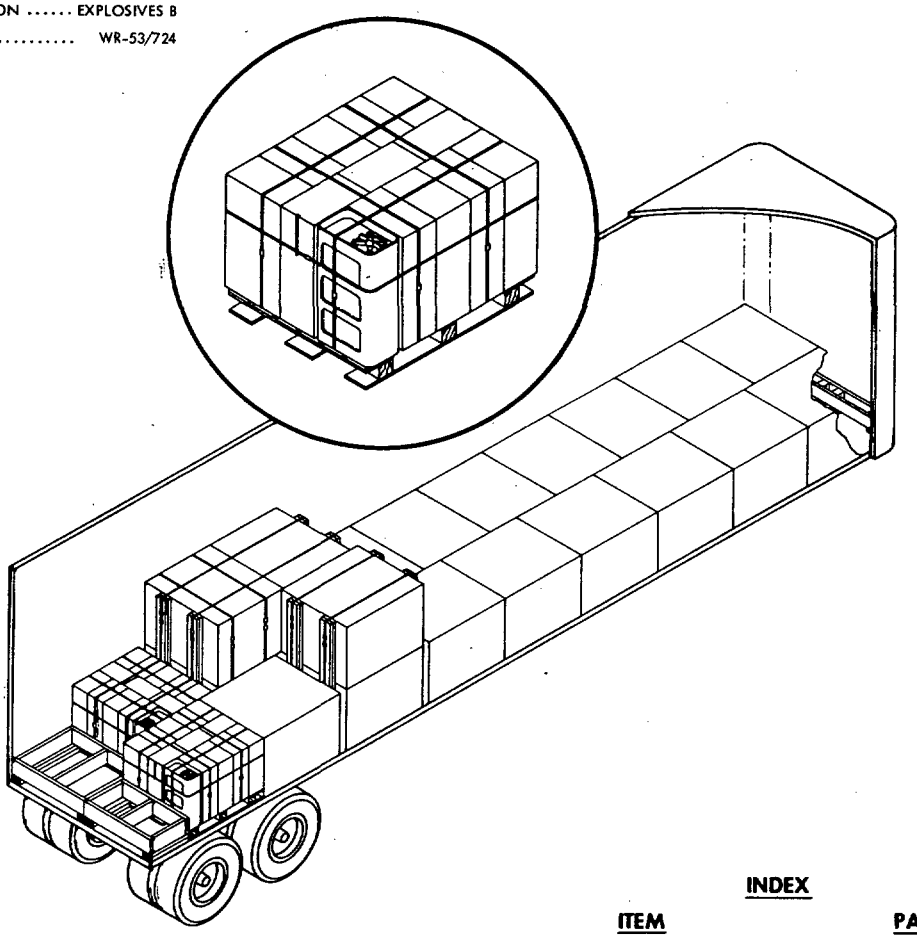
MIL-STD-1320-102
(NAVY)

2 NOVEMBER 1976

TRUCKLOADING
SMOKELESS POWDER FOR CANNON
IN PACKING BOX MK 7 AND MODS
DOMESTIC UNIT LOAD

UNIT LOAD DATA

LENGTH	48 3/4
WIDTH	42 1/4
HEIGHT	33 7/8
WEIGHT (APPROX)	2098 LBS
CUBE	40.4 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES B
UNIT LOAD DWG	WR-53/724



INDEX

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LTL & FTL	2, 5 & 6
LOAD PATTERNS	3 & 4
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NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Gray NWHC 10/29/76
TECHNICAL DIRECTION AGENT (TDA) DATE

SEASYSCOM BY DIRECTION 10/29/76 DATE

ORIGINATOR S.R. Petrina 10/29/76 DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING SMOKELESS POWDER FOR CANON IN PACKING BOX MK 7 AND MODS WHEN PALLETIZED IN ACCORDANCE WITH WR-53/724.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS. TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND LAYER WHERE THE STACK CHANGES FROM TWO LAYERS TO ONE LAYER HIGH, THE UNIT LOADS SHALL BE UNITIZED AS SHOWN ON PAGE 7. THE STIFFENERS SHALL BE POSITIONED ON THE UNIT LOADS SO THAT WHEN THE UNIT LOADS ARE IN THE TRAILER, THE STRAPPING IS PARALLEL TO THE LONGITUDINAL AXIS OF THE TRAILER. THE STIFFENERS SHALL ALSO BE POSITIONED TOWARDS THE UNIT LOADS THAT ARE ONE HIGH. IF BOTH ENDS OF THE STACKED LOADS ARE AGAINST ONE HIGH UNIT LOADS, STIFFENERS SHALL BE USED ON BOTH ENDS OF THE STACKED TWO HIGH LOADS.
4. TO PREVENT LATERAL MOVEMENT IN THE SECOND LAYER WHERE THERE IS ONLY ONE UNIT LOAD IN THE SECOND LAYER OF THE STACK, THE STACKED UNIT LOADS SHALL BE SECURED WITH TWO ADDITIONAL 1 1/4 X .035 STEEL STRAPS. THESE STRAPS ENCIRCLE THE LOADS PASSING OVER THE TOP OF THE TOP UNIT LOAD AND UNDER THE DECK OF THE PALLET OF THE BOTTOM UNIT LOAD.
5. THE REAR BLOCKING ASSEMBLIES DETAILS C AND D SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVE B" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL AND LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 21 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLES II, III, IV AND V ARE MET.

1. TABLES II, III, IV & V PROVIDE THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING USED AND THE NUMBER OF UNIT LOADS BEING LOADED INTO THE TRAILER.
2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE, SELECT THE CORRECT TABLE FOR THE LENGTH OF THE TRAILER BEING LOADED. OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED, FIND THE CORRECT LOAD PATTERN. MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT OF THE TRAILER BEING LOADED AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED.

NOTE

LIGHT WEIGHT TRACTOR-TRAILERS - THE APPROXIMATE TARE WEIGHT OF THE TRAILER TANDEM AXLES IS 7,200 LBS AND OF THE DRIVE TANDEM AXLES 10,500 LBS.

STANDARD WEIGHT TRACTOR-TRAILERS - THE APPROXIMATE TARE WEIGHT OF THE TRAILER TANDEM AXLES IS 7,800 LBS AND OF THE DRIVE TANDEM AXLES 11,500 LBS.

THE ABOVE WEIGHTS MAY BE ADJUSTED UPWARDS WHEN EXPERIENCE INDICATES THAT WEIGHT RESTRICTIONS CAN BE MET.

3. BLOCK AND BRACE THE SELECTED LOAD PATTERNS USING THE PRINCIPLES SHOWN ON PAGES 5 AND 6.
4. THE TRUCKLOADS SHOWN ON PAGES 5 AND 6 ARE PRESENTED TO SHOW THE BASIC PRINCIPLES OF LOADING. THEY DO NOT LIMIT LOADS TO THESE QUANTITIES.
5. EXCEPT WHERE OTHERWISE SPECIFIED, THE FRONT BULKHEAD, DETAIL A, IS USED FOR ALL LOADS. CERTAIN LTL LOADS THAT ARE MARKED WITH ONE, TWO OR THREE SHINGLES (#) REQUIRE THE FRONT BULKHEAD, DETAIL E TO PREVENT AXLE OVERWEIGHT.
6. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN THE LOAD PATTERN TABLES. FOLLOW THE PRINCIPLES OF THE LOAD SHOWN ON PAGE 6, USING THE FRONT BULKHEAD, DETAIL A.

TABLE I
TRUCKLOAD WEIGHT

NO. OF UNIT LOADS	TRUCKLOAD APPROX WT (LBS)
23	48,254
22	46,156
21	44,058
20	41,960
19	39,862
18	37,764
17	35,666
16	33,568

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TABLE II

40 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN FWD →	NO. OF UNIT LOADS	LOAD PATTERN FWD →	NO. OF UNIT LOADS	LOAD PATTERN FWD →
23**		21		19	
22*		20		18 OR LESS:	CONTINUE DECREASING THE NUMBER OF UNIT LOADS USING BASIC PRINCIPLES SHOWN FOR 19 UNIT LOADS USING THE STANDARD DEPTH BULKHEAD, DETAIL A, PAGE 4.

TABLE III

42 FT TRAILER

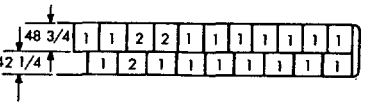
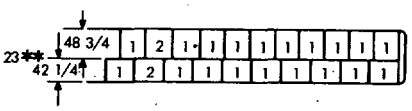
23**		20 ###		17 ###	
22*		19 ###		16 ###	
21		18 ###		15 OR LESS:	CONTINUE DECREASING THE NUMBER OF UNIT LOADS AS SHOWN ABOVE USING THE STANDARD DEPTH BULKHEAD, DETAIL A, PAGE 4.

TABLE IV

44 FT TRAILER

23**		20 ##		17 ##	
22*		19 ##		16 ##	
21 ##		18 ##		15 OR LESS:	CONTINUE DECREASING THE NUMBER OF UNIT LOADS AS SHOWN ABOVE USING THE STANDARD DEPTH BULKHEAD, DETAIL A, PAGE 3.

** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED.

* AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED.

USE 20 INCH BULKHEAD (SEE DETAIL E, PAGE 6)

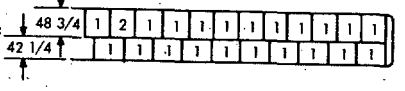
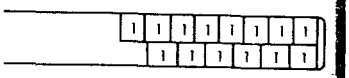
USE 25 INCH BULKHEAD (SEE DETAIL E, PAGE 6)

NOTE: WHEN SHIPPING 21 OR LESS UNIT LOADS, BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL 50 STATES WITHOUT EXCEEDING THE WEIGHT LAWS.

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TABLE V

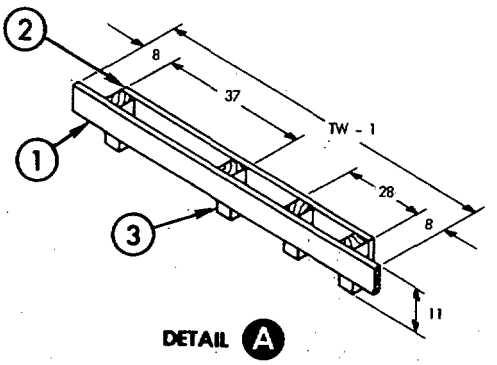
45 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN	NO. OF UNIT LOADS	LOAD PATTERN	NO. OF UNIT LOADS	LOAD PATTERN
23**		19 #		15 #	
22*		18 #		14 #	
21 #		17 #		13 OR LESS:	CONTINUE DECREASING THE NUMBER OF UNIT LOADS AS SHOWN ABOVE USING THE STANDARD DEPTH BULKHEAD, DETAIL A.
20 #		16 #			

** SEE PAGE 3

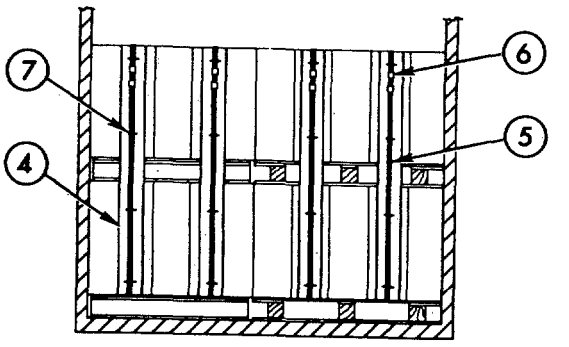
* SEE PAGE 3

USE 30 INCH BULKHEAD (SEE DETAIL E, PAGE 4)



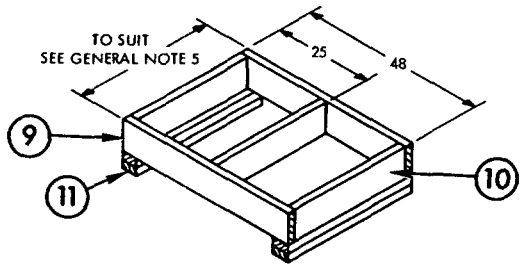
DETAIL A

FRONT BULKHEAD

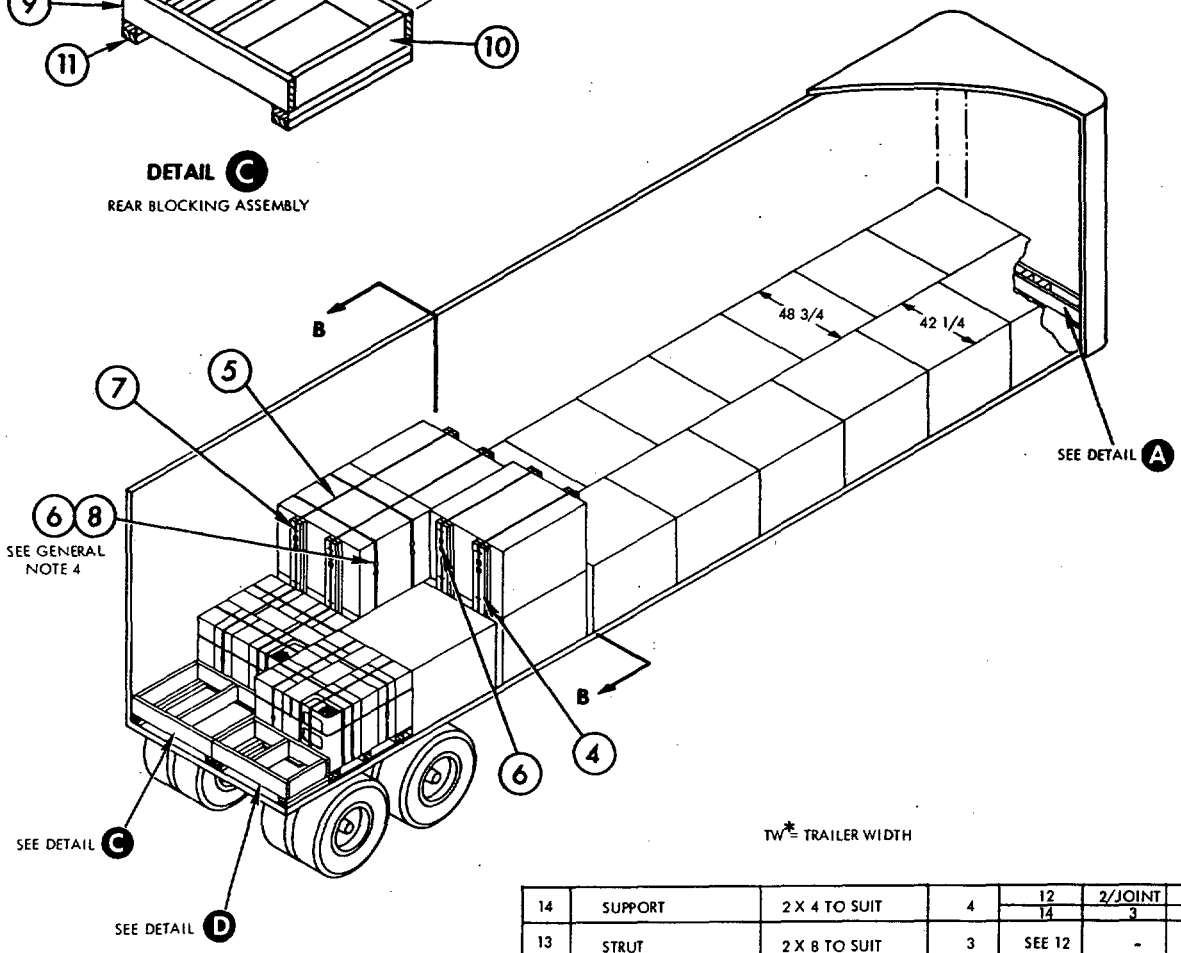


SECTION B-B

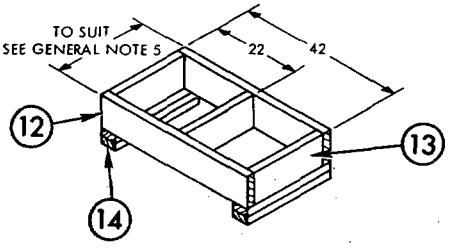
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DETAIL C
REAR BLOCKING ASSEMBLY



TW* = TRAILER WIDTH



DETAIL D
REAR BLOCKING ASSEMBLY

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
14	SUPPORT	2 X 4 TO SUIT	4	12 14	2/JOINT 3	16d 10d
13	STRUT	2 X 8 TO SUIT	3	SEE 12	-	-
12	CROSSMEMBER	2 X 8 X 42	2	13	3/JOINT	16d
11	SUPPORT	2 X 4 TO SUIT	4	9 11	2/JOINT 3	16d 10d
10	STRUT	2 X 8 TO SUIT	3	SEE 9	-	-
9	CROSSMEMBER	2 X 8 X 48	2	10	3/JOINT	16d
8	LATERAL STRAP	1 1/4 X .035 X 20 FT	2	-	-	-
7	STAPLE	FOR 1 1/4 STRAP	32	4	-	-
6	SEAL	FOR 1 1/4 STRAP	16	-	-	-
5	STRAP	1 1/4 X .035 X 20 FT	6	-	-	-
4	STIFFENER	2 X 6 X 62 1/2	16	LAMINATE (DOUBLED)	5	10d
3	UPRIGHT	4 X 4 X 11	4	SEE 1 & 2	-	-
2	FWD CROSSMEMBER	2 X 6 TO SUIT	1	3	3/JOINT	10d
1	AFT CROSSMEMBER	2 X 6 X TW* 1	1	3	3/JOINT	10d

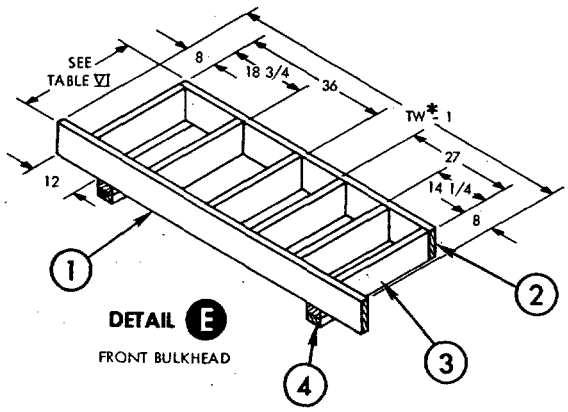
LIST OF MATERIALS & NAILING DATA

22 UNIT LOADS, 40 FT TRAILER (VAN)

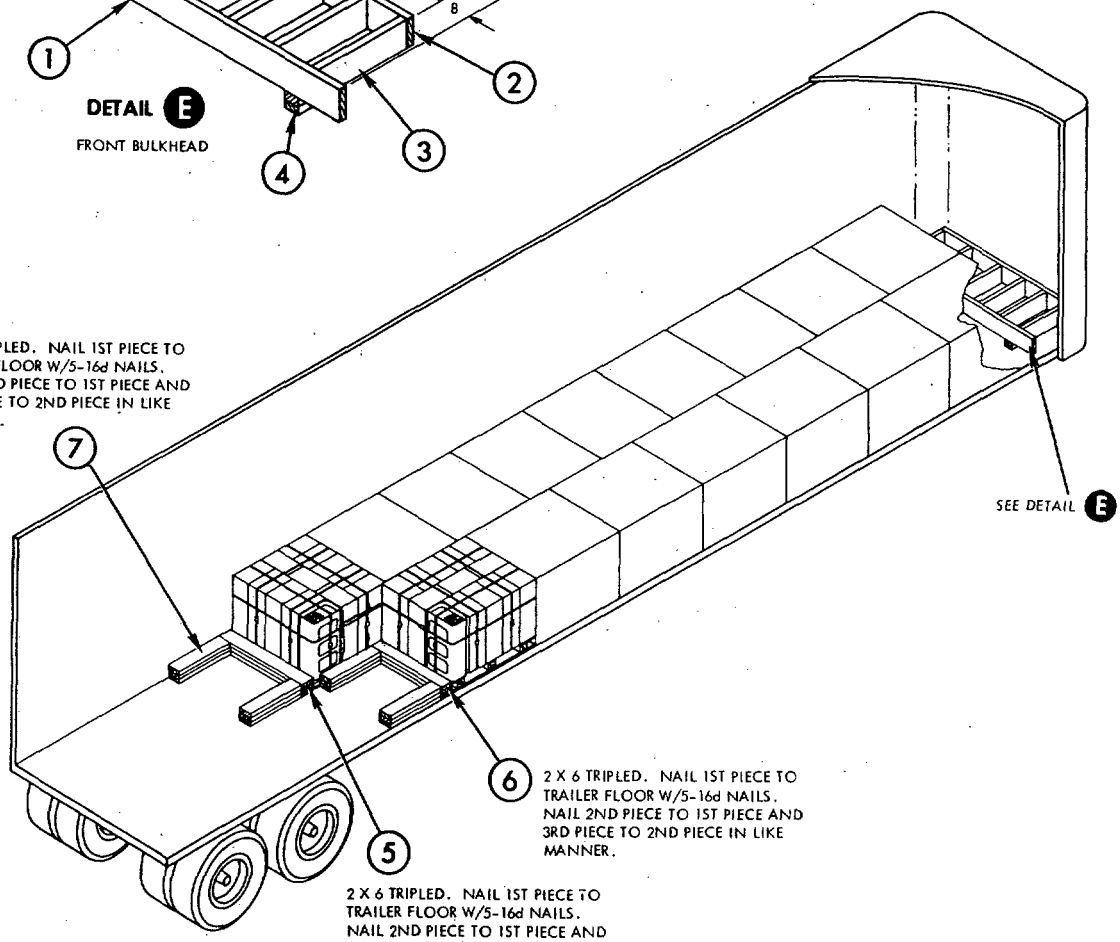
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TABLE VI

TRAILER LENGTH	BULKHEAD DEPTH
42 FT	20
44 FT	25
45 FT	30



DETAIL E
FRONT BULKHEAD



2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/5-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/5-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/5-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

TW* TRAILER WIDTH

NOTE

DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS WHEN BLOCKING BY THIS METHOD

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
7	BACKUP	2 X 6 X 30	12	SEE FIELD NOTE		
6	CROSSMEMBER	2 X 6 X 42	3	SEE FIELD NOTE		
5	CROSSMEMBER	2 X 6 X 48	3	SEE FIELD NOTE		
4	SUPPORT	2 X 4 TO SUIT	4	1 & 2 4	2/JOINT 4	16d 10d
3	STRUT	2 X 8 AS REQ'D	6	SEE 1 & 2	-	-
2	AFT CROSSMEMBER	2 X 8 TO SUIT	1	3	3/JOINT	16d
1	FWD CROSSMEMBER	2 X 8 X TW* 1	1	3	3/JOINT	16d

LIST OF MATERIALS & NAILING DATA

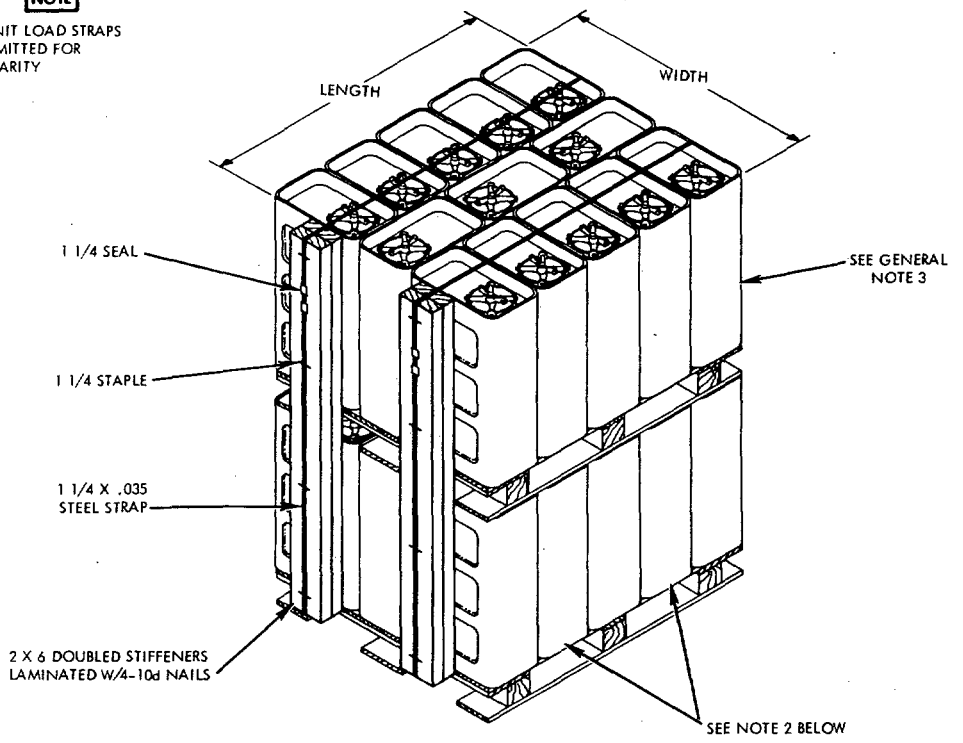
LTL OF 16 UNIT LOADS, 42, 44 & 45 FT TRAILER (VAN)

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UNITIZING PROCEDURE

NOTE

UNIT LOAD STRAPS OMITTED FOR CLARITY



1. WHEN REQUIRED BY THE FLT OR LTL REQUIREMENTS OF THIS MIL-STD, THE STACKED UNIT LOADS SHALL BE UNITIZED.

WARNING

THE LOAD PATTERN REQUIRES UNITIZING IN TWO CONFIGURATIONS: WITH STIFFENERS ON THE "WIDTH" SIDE OF THE STACKED UNIT LOADS AS SHOWN AND STIFFENERS ON THE "LENGTH" SIDE. UNITIZING SHALL BE ACCOMPLISHED SO THAT THE REQUIREMENTS OF GENERAL NOTE 3 IS SATISFIED.

2. WHEN THE STIFFENERS ARE REQUIRED TO BE ON THE "LENGTH" SIDE OF THE UNIT LOADS, POSITION EACH STIFFENER OVER THE PACKING BOX THAT IS SECOND FROM THE END.
3. THE DOUBLED 2 X 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD.
4. THE 1 1/4 INCH STEEL STRAP POSITIONED AS SHOWN, ENCIRCLE THE STACKED UNIT LOADS PASSING UNDER THE TOP DECK OF THE BOTTOM PALLET.
5. TENSION STRAPS AND SEAL WITH TWO DOUBLE NOTCHED OR CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STRAP STAPLES.

REVIEW ACTIVITY:

NAVY - OS

PREPARING ACTIVITY:

NAVY - OS
(PROJECT NO. 8140-N347)

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